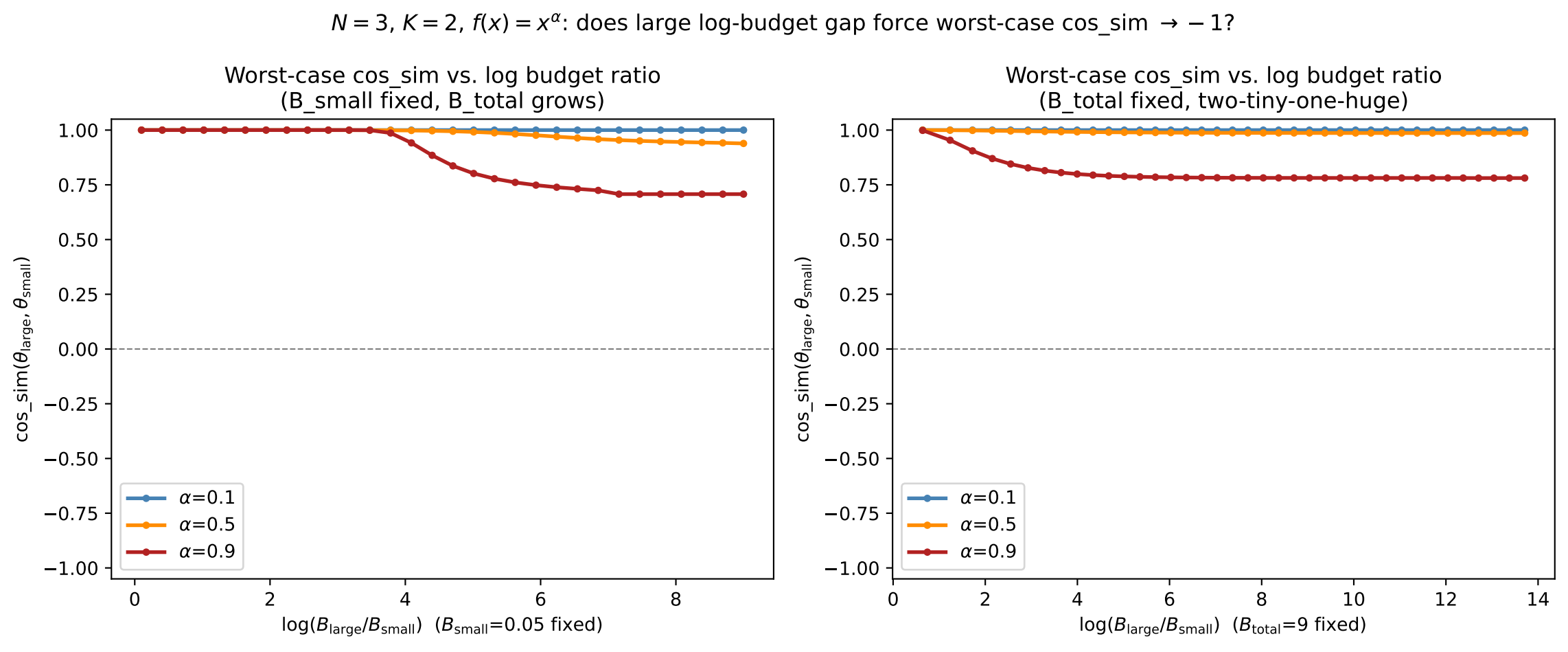
$N = 3$, $K = 2$, $f(x) = x^{\alpha}$: does large log-budget gap force worst-case cos_sim $\to -1$?
Worst-case cos_sim vs. log budget ratio
(B_small fixed, B_total grows)
cos_sim(θ_{large}, θ_{small})
1.00
0.75
0.50
0.25
0.00
−0.25
−0.50
−0.75
−1.00
0
2
4
6
8
log(B_{large}/B_{small}) (B_{small}=0.05 fixed)
α=0.1
α=0.5
α=0.9
Worst-case cos_sim vs. log budget ratio
(B_total fixed, two-tiny-one-huge)
cos_sim(θ_{large}, θ_{small})
1.00
0.75
0.50
0.25
0.00
−0.25
−0.50
−0.75
−1.00
0
2
4
6
8
10
12
14
log(B_{large}/B_{small}) (B_{total}=9 fixed)
α=0.1
α=0.5
α=0.9